

ENGINEERING PORTFOLIO

Rishabh Bavithran

Designed CarbonBridge's core reactor technology from first principles - then automated it into a product sold to two research institutions.

16×

YIELD

2

SOLD TO

DGF-100

FLAGSHIP

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The DGF-100 - dual-reactor system, CarbonBridge Inc.

The Short Version

What I built, why it mattered, and what happened next.

- Designed the physical reactor, fluid paths, and a patent-pending microbe-delivery method - from scratch
- Proved the design by de-risking: manual valve testing and delivery methods before writing any control software
- Authored the full C++ automation stack that turned it into the **DGF-100**, a sellable product
- Sold to Lawrence Berkeley National Lab and A*STAR Singapore (delivery pending) - running inside CarbonBridge's own lab



16×

METHANOL YIELD
(8 MG → 128 MG)

2

INSTITUTIONS SOLD TO
LBNL • A*STAR

4

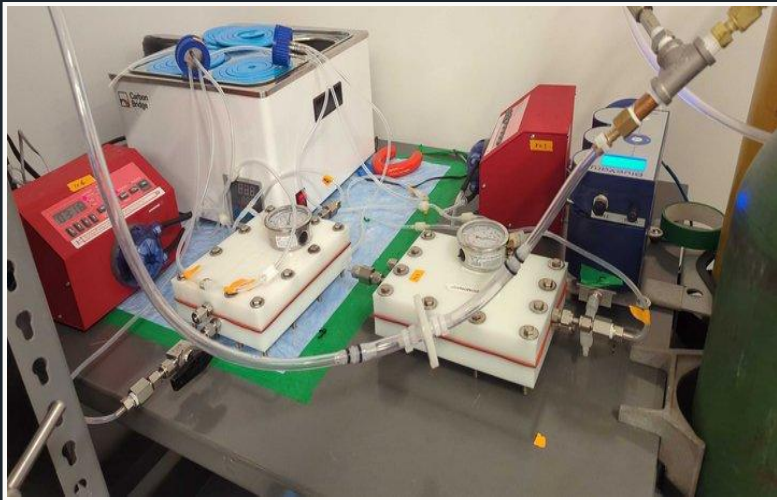
MAJOR BUILD
ITERATIONS

*Every stage on this presentation fed into one thing:
the DGF-100 that exists today.*

The Build Journey

Three stages, each stage de-risked before the next was built - from a 10 cm² lab component to a purpose-built test rig.

01 • ORIGINS (FEB-APR 2025)



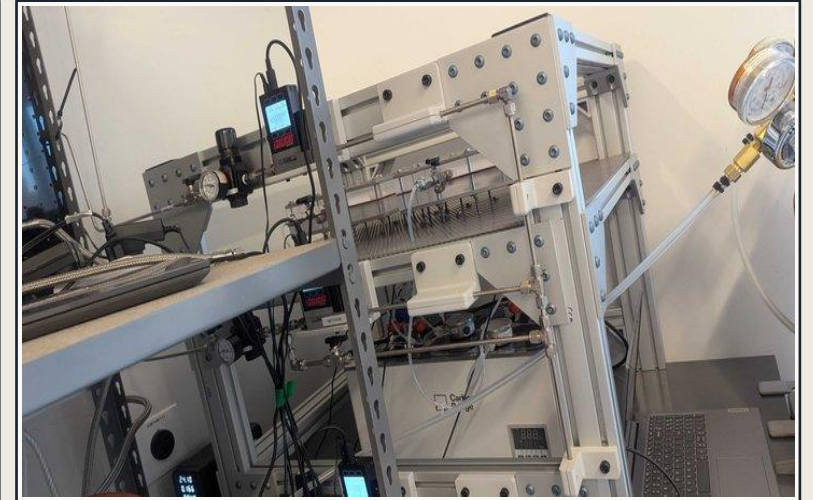
Mini Notebook V1 - first scale-up from the original 10 cm² component; gas delivery and leak checks required manual, hands-on intervention

02 • DE-RISKING (APR-AUG 2025)



Full-size Notebook at 4× scale, run with manual leak checks, microbe delivery and gas delivery

02 • REACTOR SHELF (AUG-OCT 2025)

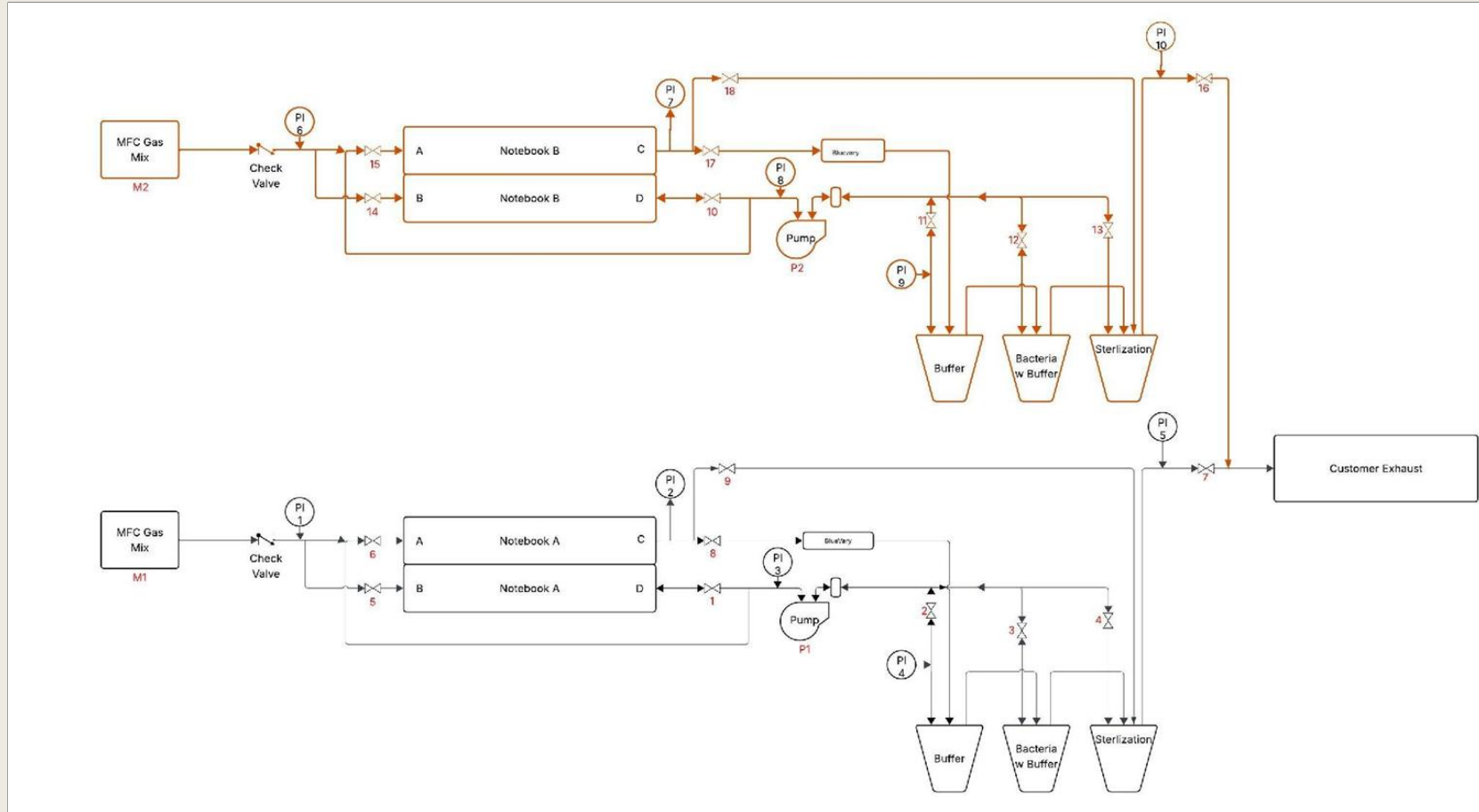


A purpose-built rack I designed to organize both reactors for systematic manual testing

A detour that didn't ship: also built a stainless-steel mini notebook - it worked, but needed a different control architecture, so I reverted to keep software consistent across scale.

The Architecture Behind It

Before any code was written, every valve and gas path for both reactors was mapped out here - this P&ID became the logic the automation software runs on today.



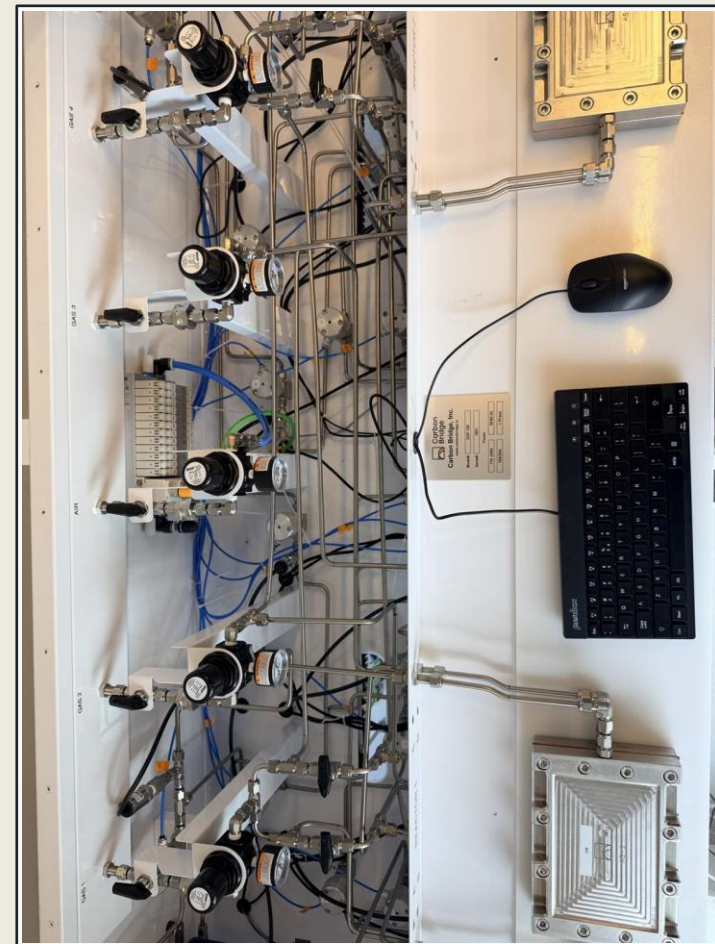
The DGF-100



The full DGF-100 - both reactors, gas manifold, and control software

Everything above became a fully automated, dual-reactor system - sold to two research institutions.

- Complete C++ automation stack, authored from scratch
- Modbus TCP/RTU and RS-232 hardware abstraction layers
- Concurrent safety-monitoring threads + retry/recovery logic
- JSON-driven experiment and test configs - no rebuild for new protocols
- Full electromechanical build: wiring, tubing, assembly



Behind the panel - the full gas manifold and wiring for four independent, labeled gas lines

Beyond the DGF-100

Outside the flagship build, I support the science team's day-to-day infrastructure:

- Methane sensor for autoclave gas fills - mounting, tubing, wiring, and readout, designed and built end to end
- Nursery gas delivery & DAQ-based temperature monitoring - thermocouples tracking culture-sensitive thermal conditions



Methane sensor, inline



Nursery + DAQ monitoring

SKILLS

Mechanical: SolidWorks, Fusion360, CREO, P&ID Design, CNC hardware, gas/fluid systems

Electronics: Modbus, RS-232/485, I2C/SPI/UART, DAQ, microcontrollers

Software: C++, Python, ROS2, Git/Tailscale/SSH, JSON-driven configs

LET'S TALK

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The pattern underneath all of this: test something small, learn from it, fold that in, and go again - at every scale, every time.